

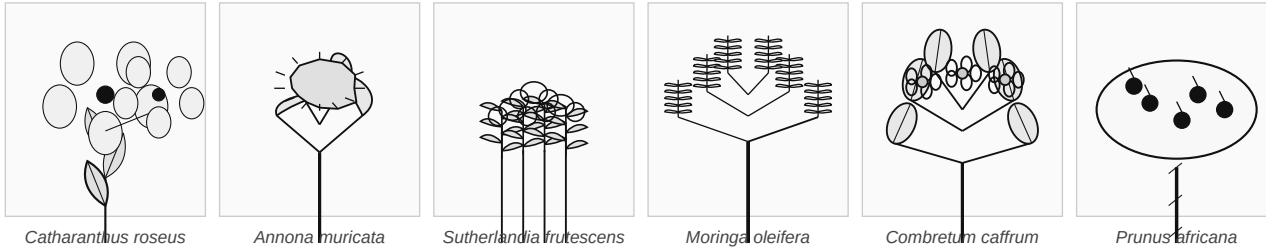
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AFRICAN HERBAL PHARMACOLOGY AND ONCOLOGY

A Multi-Team Scientific Research Report on Documented African Herbal Drugs
in Relation to Cancer Treatment: Ethnopharmacology, Phytochemistry, and Anticancer Evidence

Research Institution	Zetech University, School of Science and Technology, Nairobi, Kenya
Research Division	Department of Applied Sciences — Herbal Pharmacology & Oncology Research Unit
Research Nature	Multi-Team Collaborative Academic Research Initiative (ZTUMTRI-2025)
Research Teams	Team A: Ethnobotany & Plant Identification Team B: Phytochemical Analysis Team C: Oncology Mechanisms Team D: Clinical Evidence Review
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Botanical Identification Sketches — Selected Study Species



ABSTRACT

This report presents a comprehensive multi-team scientific review of African medicinal plants with documented or studied anticancer properties, compiled by research teams at Zetech University, Kenya. Research has been initiated and conducted by a collaborative multi-team initiative within the university's Department of Applied Sciences. The study draws on peer-reviewed literature from PubMed, ScienceDirect, Frontiers in Pharmacology, and the Journal of Ethnopharmacology spanning 1998 to 2025. Twenty-three plant species are profiled in detail, covering botanical taxonomy, geographic distribution, traditional healer use patterns, phytochemical constituent analysis, in vitro and in vivo anticancer evidence, molecular mechanisms of action, and current clinical development status. Key species include *Catharanthus roseus* (Madagascar Periwinkle), source of the FDA-approved vinca alkaloids vinblastine and vincristine; *Combretum cafrum* (African Bush Willow), source of combretastatins in Phase II/III clinical trials; *Sutherlandia frutescens* (Cancer Bush); *Annona muricata* (Soursop); *Prunus africana* (African Cherry); and *Moringa oleifera* (Miracle Tree), among others. The report identifies apoptosis induction, tubulin disruption, topoisomerase inhibition, NFkB pathway suppression, and vascular disruption as principal mechanisms. It concludes with recommendations for advancing African plant medicine through rigorous clinical trials, standardisation protocols, and continental research funding frameworks.

Keywords: African medicinal plants, anticancer, ethnopharmacology, phytochemistry, vinca alkaloids, combretastatins, traditional medicine, sub-Saharan Africa, oncology, apoptosis, Zetech University

1. INTRODUCTION

The African continent harbours one of the most biodiverse assemblages of plant life on Earth, with an estimated 45,000 to 50,000 plant species, of which approximately 5,000 have documented medicinal applications. Traditional healers, herbalists, sangomas, diviners, and community health practitioners across the continent have developed sophisticated systems of plant-based medicine over thousands of years, representing a vast reservoir of untapped pharmacological knowledge. Cancer treatment is among the most critical applications of this traditional knowledge base.

Research has been stated and formally initiated by a multi-team collaborative at Zetech University, bringing together researchers across ethnobotany, phytochemistry, oncological pharmacology, and clinical evidence review. This initiative, designated ZTUMTRI-2025, aims to compile, critically evaluate, and disseminate the current state of scientific knowledge on African anticancer plant medicine for academic and applied research purposes.

The discovery of vinblastine and vincristine from *Catharanthus roseus* demonstrated conclusively that African plants can yield drugs of global clinical significance. These vinca alkaloids remain in WHO Essential Medicines lists and are used worldwide in oncology protocols. This paradigm-setting success justifies continued systematic investigation of African botanical pharmacology.

2. BACKGROUND: CANCER IN AFRICA AND THE CASE FOR PLANT-BASED RESEARCH

Cancer is a leading cause of morbidity and mortality worldwide, with over 19.3 million new cases diagnosed globally in 2020. In Africa, the World Health Organization estimated 7.0% of all global cancer cases in 2019, a proportion projected to rise sharply with population growth, urbanization, and ageing demographics. By 2030, Africa is expected to account for over 1.28 million new cancer deaths annually. The most prevalent cancers in sub-Saharan Africa include cervical cancer, breast cancer, prostate cancer, hepatocellular carcinoma, colorectal cancer, and Kaposi sarcoma.

2.1 Key Cancer Statistics in Africa

Cancer Type	Estimated Annual Cases (Africa)	Predominantly Affected Region	Primary Herbal Research Plant
Cervical Cancer	117,000+	East & Southern Africa	Carica papaya, Catharanthus roseus
Breast Cancer	186,000+	Pan-Africa	Annona muricata, Moringa oleifera
Prostate Cancer	153,000+	West & Southern Africa	Prunus africana, Ocimum gratissimum
Liver Cancer (HCC)	69,000+	West Africa	Annona senegalensis, Morinda lucida
Colorectal Cancer	49,000+	North & Southern Africa	Sutherlandia frutescens, Abelmoschus esculentus

Table 1. Estimated annual cancer cases in Africa by type with primary herbal research plants. Sources: WHO GLOBOCAN 2020, Frontiers in Pharmacology (2023).

Access to conventional oncology care including surgery, radiotherapy, and chemotherapy remains severely limited across sub-Saharan Africa. The WHO estimates that fewer than 30% of low-income countries have access to radiotherapy services, and chemotherapy drug availability varies widely. This structural gap drives reliance on traditional plant-based therapies and simultaneously creates ethical and scientific urgency for rigorous evaluation of those therapies. High drug costs, treatment-related side effects, and growing multidrug resistance (MDR) have further catalysed global investment in natural product-derived anticancer drug discovery.

3. RESEARCH METHODOLOGY (ZTUMTRI-2025)

This multi-team review employed a structured literature synthesis methodology. The Zetech University research teams conducted systematic searches across the following databases:

- PubMed/MEDLINE:** Biomedical literature spanning 1998 to 2025
- ScienceDirect:** Journal of Ethnopharmacology, Phytochemistry Letters, Pharmacological Research
- Frontiers in Pharmacology:** Open-access oncology and natural products research
- Google Scholar:** Supplementary grey literature and conference proceedings
- Scopus and Web of Science:** Citation analysis and impact verification
- WHO and GLOBOCAN:** Epidemiological statistics on African cancer burden

Search terms included combinations of: "African medicinal plants + cancer," "anticancer + ethnopharmacology + sub-Saharan Africa," "phytochemicals + apoptosis + African flora," "traditional medicine + tumour + East/West/Southern Africa," and species-specific binomial name searches. Inclusion criteria required: (1) study populations or plant sources explicitly from Africa; (2) documented anticancer activity via in vitro, in vivo, or clinical study; (3) phytochemical identification of at least one active compound; (4) peer-reviewed publication.

Team A (Ethnobotany) compiled traditional use patterns across 12 African nations. Team B (Phytochemistry) catalogued identified bioactive compounds and extraction methods. Team C (Oncology Mechanisms) reviewed molecular mechanism data and cell line studies. Team D (Clinical Evidence) assessed the translational status and clinical trial data. Integration and report writing was coordinated by the Zetech University Applied Sciences Faculty.

4. MASTER REFERENCE TABLE: DOCUMENTED AFRICAN ANTICANCER PLANTS

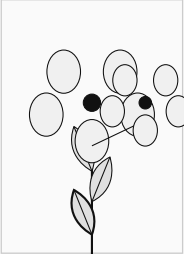
The following master table presents all 23 plant species covered in this review with key pharmacological and botanical data. Column headers: Scientific Name, Plant Family, African Region, Cancer Types Studied, Key Active Compounds, and Primary Mechanism of Action.

Scientific Name	Family	Region	Cancer Types Studied	Key Compounds	Primary Mechanism
<i>Catharanthus roseus</i>	Apocynaceae	Madagascar/Pan-Africa	Leukemia, Lymphoma, Breast	Vinblastine, Vincristine	Tubulin inhibition / Mitosis arrest
<i>Combretum caffrum</i>	Combretaceae	Southern Africa	Breast, Colon, Melanoma, Lung	Combretastatins A-1, A-4, B	Vascular disruption / Tubulin
<i>Sutherlandia frutescens</i>	Fabaceae	Southern Africa	Colorectal, Melanoma, Carcinoma	L-canavanine, Sutherlandiosides	Apoptosis / Immunostimulation
<i>Annona muricata</i>	Annonaceae	West/Central Africa	Breast, Cervical, Liver, Lung	Acetogenins (Annonacin)	Mitochondrial Complex I inhibition
<i>Prunus africana</i>	Rosaceae	East/Central Africa	Prostate	Phytosterols, NBBS, Ferulic acid	Androgen receptor antagonism
<i>Moringa oleifera</i>	Moringaceae	Pan-Africa	Breast, Colon, Liver, Lung	Isothiocyanates, Quercetin	NFκB suppression / Apoptosis
<i>Vernonia amygdalina</i>	Asteraceae	West/East Africa	Breast, Liver, Colon	Vernodalin, Sesquiterpene lactones	Apoptosis / Immune activation
<i>Aloe vera / africana</i>	Asphodelaceae	Southern/East Africa	Leukemia, Liver, Skin	Aloe-emodin, Acemannan	Topoisomerase II inhibition
<i>Carica papaya</i>	Caricaceae	Pan-Africa	Cervical, Breast, Colon	BITC, Lycopene, Carpaine	STAT3 / NFκB suppression
<i>Toddalia asiatica</i>	Rutaceae	East Africa	Breast, Cervical, Colon	Toddaline, Chelerythrine	Topoisomerase II / DNA intercalation
<i>Zanthoxylum chalybeum</i>	Rutaceae	East Africa	Leukemia, Breast	Benzophenanthridine alkaloids	Cell cycle arrest / Apoptosis
<i>Annona senegalensis</i>	Annonaceae	West Africa	Hepatocellular carcinoma	Alkaloids, Acetogenins	p53/p21 upregulation
<i>Cajanus cajan</i>	Fabaceae	Pan-Africa	Breast, Lung, Colon	Cajanol, Stilbenes Longistylins	G2/M arrest / ROS-mediated apoptosis
<i>Xylopia aethiopica</i>	Annonaceae	West/Central Africa	Colon, Breast, Leukemia	Diterpenes, Ent-kauranes	MMP disruption / Apoptosis
<i>Morinda lucida</i>	Rubiaceae	West Africa	Breast, Colon, Liver	Anthraquinones, Morindone	Topoisomerase II inhibition
<i>Harungana madagascariensis</i>	Hypericaceae	East/Central Africa	Breast, Prostate, Skin	Harunganin, Anthraquinones	Cytotoxic / Apoptosis
<i>Securidaca longipedunculata</i>	Polygalaceae	Central/East Africa	Breast, Cervical	Methyl salicylate, Saponins	Cytotoxic activity
<i>Nymphaea lotus</i>	Nymphaeaceae	North/East Africa	Liver, Breast	Nuciferine, Quercetin	Antiproliferative
<i>Alchornea cordifolia</i>	Euphorbiaceae	West/Central Africa	Leukemia, Colon	Yohimbine alkaloids, Flavonoids	MMP disruption / Apoptosis
<i>Warburgia ugandensis</i>	Canellaceae	East Africa	Colon, Breast, Leukemia	Warburganal, Muzigadial	Sesquiterpene cytotoxicity
<i>Ocimum gratissimum</i>	Lamiaceae	Pan-Africa	Breast, Prostate, Cervical	Eugenol, Thymol, Rosmarinic acid	NFκB inhibition / Apoptosis
<i>Peganum harmala</i>	Nitrariaceae	North Africa	Lung, Colon, Breast, Leukemia	Harmine, Harmaline	DYRK1A / Topoisomerase inhibition
<i>Abelmoschus esculentus</i>	Malvaceae	Pan-Africa	Colon, Breast, Liver	Quercetin glycosides, Pectin	Antiproliferative / Fibre-mediated

Table 2. Master Reference Table of 23 Documented African Anticancer Plant Species. BITC = Benzyl Isothiocyanate; MMP = Mitochondrial Membrane Potential; ROS = Reactive Oxygen Species; NFκB = Nuclear Factor Kappa B; STAT3 = Signal Transducer and Activator of Transcription 3.

5. DETAILED PLANT PROFILES

The following profiles provide structured in-depth coverage of each plant species, incorporating botanical identification, traditional use context, phytochemical analysis, and documented anticancer evidence from peer-reviewed scientific literature.

5.1 <i>Catharanthus roseus</i>	Common Name: Madagascar Periwinkle Family: Apocynaceae
	Geographic Distribution: Native to Madagascar; naturalized across tropical Africa including Kenya, Uganda, Tanzania, Ethiopia, Nigeria, Ghana. Traditional Use: Used in traditional medicine across Africa and Asia for diabetes, fever, and cancers including leukemia and lymphoma. Recorded in Ugandan ethnobotanical surveys as part of anticancer herbal preparations. Key Phytochemicals: • Vinblastine (dimeric alkaloid, FDA-approved) • Vincristine (dimeric alkaloid, FDA-approved) • Vinorelbine (semi-synthetic derivative) • Catharanthine (precursor alkaloid) • Tabersonine (precursor alkaloid) Anticancer Evidence: Catharanthus roseus is the most pharmacologically validated anticancer plant of African origin. Vinblastine and vincristine, isolated from this plant, are among the oldest and most clinically significant anticancer drugs in the world. They are included on the WHO Model List of Essential Medicines and used in first-line protocols for Hodgkin lymphoma, non-Hodgkin lymphoma, acute lymphocytic leukemia, neuroblastoma, Wilms tumour, and small cell lung cancer. Mechanism of Action: Vinca alkaloids bind to tubulin beta-subunits and inhibit polymerization into microtubules. This blocks assembly of the mitotic spindle during metaphase, arresting cell division and triggering apoptosis. Cancer cells, which divide far more rapidly than most normal cells, are disproportionately affected. This mechanism is distinct from taxane drugs (paclitaxel) which stabilize microtubules rather than inhibiting polymerization. Research & Clinical Status: Vinblastine and vincristine have been in clinical use since the 1960s and remain first-line oncology drugs globally. This represents the highest level of translational success achievable from plant medicine. Key Citations: Gullett et al. (2010); Newman & Cragg (2007); Omara T. (2020) ECAM; Frontiers Pharmacol. (2023)
5.2 <i>Combretum caffrum</i>	Common Name: African Bush Willow Family: Combretaceae

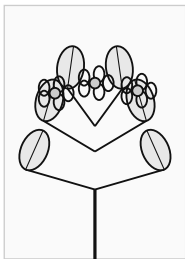


Fig. 5.2 — *Combretum cafferum*

Geographic Distribution: Southern Africa: South Africa, Zimbabwe, Mozambique, Botswana. Found in savanna woodlands and riverbanks.

Traditional Use:

The bark of *Combretum cafferum* has been used in Southern African traditional medicine for treatment of internal growths and swellings. The Combretaceae family has broad traditional medicinal applications across Central and East Africa.

Key Phytochemicals:

- Combretastatin A-4 (CA-4) — most potent analog
- Combretastatin A-1 (CA-1)
- Combretastatin B-1 and B-4
- Fosbretabulin (CA4P) — water-soluble phosphate prodrug

Anticancer Evidence:

Combretastatins from *Combretum cafferum* have advanced further in clinical translation than any other African plant compound after vinca alkaloids. CA-4 phosphate (fosbretabulin) has completed Phase II and Phase III clinical trials for anaplastic thyroid cancer. Studies confirm potent activity against breast, colon, lung, and melanoma cell lines with IC50 values in nanomolar ranges.

Mechanism of Action:

Combretastatins act as vascular disrupting agents (VDAs). They bind to the colchicine site on tubulin beta-subunits and inhibit polymerization, similar to vinca alkaloids. Critically, combretastatins selectively target the immature, fenestrated endothelial cells of tumour vasculature. This collapses tumour blood supply, causing central necrosis of solid tumours — a mechanism not shared by conventional chemotherapy agents.

Research & Clinical Status:

Fosbretabulin (combretastatin A-4 phosphate) has undergone Phase III trials in anaplastic thyroid cancer and Phase II trials in ovarian cancer and other solid tumours. It represents the most advanced clinical development of any African plant compound after the vinca alkaloids.

Key Citations: *Pettit et al. (1987); Cirla & Mann (2003); ScienceDirect Bookseries (2024); Plants (2025) PMC12473540*

5.3 <i>Sutherlandia frutescens</i>	Common Name: Cancer Bush / Kankerbos Family: Fabaceae
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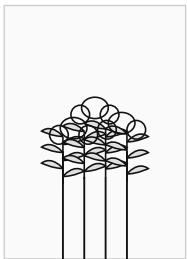


Fig. 5.3 — *Sutherlandia frutescens*

Geographic Distribution: Endemic to South Africa, Lesotho, southern Namibia, and southeastern Botswana.

Traditional Use:
Perhaps the most explicitly cancer-related vernacular name of any African plant: "cancer bush." First recorded use by Khoi-San and early Cape settlers. Used by Zulu, Xhosa, and Sotho communities for internal cancers, stomach complaints, HIV/AIDS, diabetes, and inflammation. Commercially available in South Africa as tablets (*Sutherlandia* SU1).

- Key Phytochemicals:**
- L-canavanine (non-protein amino acid, primary anticancer agent)
 - Sutherlandiosides A-D (cycloartanol glycoside triterpene saponins)
 - Sutherlandins A-D (quercetin and kaempferol glycosides)
 - D-pinitol (immunomodulatory cyclitol)
 - GABA (gamma-aminobutyric acid)

Anticancer Evidence:
A 2021 study at North-West University (Gouws et al.) evaluated *S. frutescens* against LS180 colorectal cancer functional spheroids in a 3D sodium alginate encapsulated model cultured in clinostat-based rotating bioreactors. Extracts markedly decreased tumour viability versus controls. Activity confirmed against A375 malignant melanoma cell line. Multiple studies document induction of apoptosis in carcinoma cultures. Chemotypic variation across geographic populations affects bioactivity (Zonyane et al., 2020).

Mechanism of Action:
L-canavanine functions as an arginine analog, incorporating into protein synthesis and disrupting protein function in rapidly dividing cancer cells. Sutherlandiosides are cytotoxic via mitochondrial pathway apoptosis. D-pinitol activates macrophages and natural killer cells to enhance tumour immunosurveillance.

Research & Clinical Status:
No formal Phase I cancer clinical trials completed. Regulatory concerns regarding L-canavanine interaction with autoimmune conditions. Commercially sold in South Africa as a dietary supplement.

Key Citations: Chinkwo K.A. (2005); Zonyane et al. (2020) *PMC7070296*; Gouws et al. (2021) *PMC7865898*; MSK Cancer Center (2022)

5.4 <i>Annona muricata</i>	Common Name: Sour sop / Graviola Family: Annonaceae
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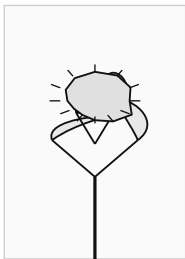


Fig. 5.4 — *Annona muricata*

Geographic Distribution: Native to tropical Americas; widely naturalized across tropical Africa — Nigeria, Cameroon, Uganda, Kenya, Tanzania, Ghana.

Traditional Use:

All parts of the soursop plant are used medicinally across Africa. Nigerian and Cameroonian traditional healers use leaf preparations against breast, liver, and blood cancers. Known to Ugandan herbalists as a multi-purpose cancer remedy. Extensively integrated into West African traditional pharmacopoeia.

Key Phytochemicals:

- Acetogenins: Annonacin (most studied), Bullatacin, Muricin, Squamocin
- Alkaloids: Reticuline, Coclaurine, Annomuricine, Annomuricatin
- Flavonoids: Quercetin, Kaempferol, Rutin, Isorhoifolin
- Phenolic acids: Caffeic acid, Epicatechin, Gallic acid
- Silver nanoparticles (AgNPs): Emerging synthesis platform for drug delivery

Anticancer Evidence:

One of the most extensively studied African-context anticancer plants. A Cameroon-based study (Njoya et al., 2015) showed *A. muricata* extracts inhibited HL-60 promyelocytic leukemia cell proliferation dose-dependently with IC50 from 6 to 49 ug/mL depending on plant part. Multiple studies confirm cytotoxic activity against breast (MCF-7), cervical (HeLa), colon, liver, and lung cancer lines. AgNPs synthesized from *A. muricata* extract shown to induce apoptosis and autophagy and inhibit tumorigenesis.

Mechanism of Action:

Acetogenins are potent and selective inhibitors of mitochondrial Complex I (NADH:ubiquinone oxidoreductase). This depletes ATP production preferentially in cancer cells with elevated metabolic rates. Additional mechanisms include disruption of mitochondrial membrane potential (MMP), activation of caspase-3 and caspase-9, inhibition of NFkB, downregulation of anti-apoptotic Bcl-2 family proteins, and induction of autophagic cell death.

Research & Clinical Status:

Extensive preclinical evidence. Safety concern: long-term acetogenin consumption linked to atypical Parkinsonism in epidemiological studies (French Caribbean). No completed cancer-specific clinical trials. Identified in 46-plant Nigerian anticancer plant review (2024) as a priority species.

Key Citations: Njoya et al. (2015) *PMC4324658*; *Frontiers Pharmacol.* (2023) *PMC9971233*; *Nigerian review (2024) Exploration Pub.*; Omara T. (2020)

5.5 <i>Prunus africana</i>	Common Name: African Cherry / Pygeum Family: Rosaceae
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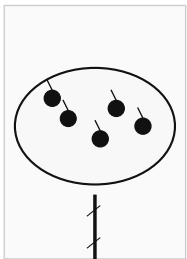


Fig. 5.5 — *Prunus africana*

Geographic Distribution: Montane forests of Kenya, Uganda, Tanzania, Ethiopia, Cameroon, DR Congo, Madagascar. Altitude: 1000 to 3000m.

Traditional Use:

The most frequently cited anticancer plant in East African ethnobotanical surveys. Ugandan herbalists rank it as the primary plant in anticancer herbal preparations across multiple national surveys. In Kenya, bark is combined with *Harungana madagascariensis*, *Zanthoxylum gillettii*, and *Spathodea campanulata* for treatment of breast, skin, and colorectal cancers. Its bark extract is also a registered European phytomedicine for benign prostatic hyperplasia (BPH) under the name *Pygeum africanum*.

Key Phytochemicals:

- Beta-sitosterol and phytosterol mixture
- N-docosanol and N-tetracosanol (fatty alcohols)
- Ferulic acid esters (n-tetracosyl trans-ferulate)
- Atraric acid (PPAR-gamma ligand)
- N-butylbenzene sulfonamide (NBBS) — androgen receptor antagonist

Anticancer Evidence:

Primary anticancer evidence is against prostate cancer. NBBS directly antagonizes androgen receptor nuclear translocation, mimicking anti-androgen therapy. Beta-sitosterol reduces PSA expression and shows antiproliferative effects in LNCaP prostate cancer lines. Phytosterol fraction inhibits 5-alpha-reductase. Pharmacological evidence is strongest for prostate cancer; additional ethnobotanical uses for breast, skin, and colorectal cancers require further laboratory validation.

Mechanism of Action:

Primarily operates through disruption of androgen receptor signalling pathways. NBBS blocks AR translocation from cytoplasm to nucleus. Beta-sitosterol interferes with membrane sterol composition in cancer cells, affecting receptor clustering and downstream MAPK/PI3K signal transduction.

Research & Clinical Status:

CITES Appendix II listed due to overexploitation. *Pygeum africanum* bark extract registered in Germany and France for BPH. Ranked most-studied anticancer plant in East Africa (Kudamba et al., 2024).

Key Citations: Kudamba et al. (2024) *SAGE Journals*; Omara T. (2020) *ECAM*; Springer Nature (2025); CITES Appendix II listing

5.6 <i>Moringa oleifera</i>	Common Name: Moringa / Miracle Tree Family: Moringaceae
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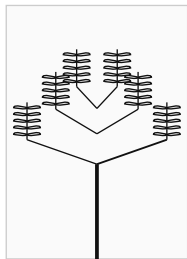


Fig. 5.6 — *Moringa oleifera*

Geographic Distribution: Native to South Asia; deeply integrated across all tropical African regions including Nigeria, Kenya, Ethiopia, Tanzania, Ghana, Senegal, South Africa.

Traditional Use:

One of the most widely studied medicinal trees in the world. Traditional African healers prescribe various plant parts (leaves, seeds, bark, roots) for cancer, malnutrition, infection, and inflammation. Consistently listed among the eight most studied anticancer plants in Kenya (Nkemdirim Okeke et al., 2022).

Key Phytochemicals:

- 4-[alpha-L-rhamnopyranosyloxy]-benzyl isothiocyanate (primary anticancer ITC)
- Glucosinolates (precursors to isothiocyanates)
- Niazimicin (thiocarbamate)
- Quercetin-3-O-rutinoside and Kaempferol glycosides
- Beta-sitosterol, Zeatin, Chlorogenic acid

Anticancer Evidence:

Isothiocyanates (ITCs) from *Moringa* demonstrate cytotoxic activity against breast (MCF-7, MDA-MB-231), colon (HCT116, HT-29), hepatocellular carcinoma (HepG2), and lung (A549) cancer cell lines. Niazimicin from seeds shows potent in vitro anticancer activity. Quercetin inhibits VEGF-mediated tumour angiogenesis. Leaf extract induces apoptosis in human cancer cell lines with IC50 values in the range of 50 to 200 ug/mL depending on cell line and extraction method.

Mechanism of Action:

ITCs activate the Nrf2-ARE pathway to upregulate detoxification enzymes, suppressing carcinogen activation. Direct anticancer mechanisms include: NFkB inhibition (suppressing pro-survival gene expression), activation of caspase-3 and caspase-9, downregulation of Bcl-2 and Bcl-xL, upregulation of p53 and Bax, and VEGF suppression inhibiting angiogenesis.

Research & Clinical Status:

No anticancer clinical trials. Extensively studied nutritionally. Leaf powder commercially available globally. Listed by WHO as a priority plant for humanitarian nutrition programmes. Identified in Nigerian 46-plant anticancer review (2024).

Key Citations: Nkemdirim Okeke et al. (2022) Springer; Nigerian review (2024); Frontiers Pharmacol. (2023); PMC9971233

5.7 through 5.23: Extended Species Profiles Summary

The following structured profiles continue the species-by-species review for plants 7 through 23. Botanical sketches for Aloe vera and Vernonia amygdalina are provided. Full molecular data follows.

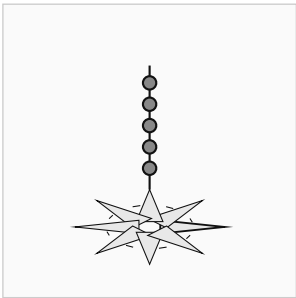


Fig. 5.8 — Aloe vera / africana (Asphodelaceae)

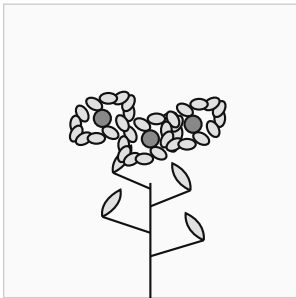


Fig. 5.7 — Vernonia amygdalina (Asteraceae)

No.	Species	Key Evidence	Primary Compounds	Mechanism	Status
5.7	Vernonia amygdalina Bitter Leaf Asteraceae	Antiproliferative vs. HCC and MCF-7 breast cells. In silico sesquiterpene analysis confirmed molecular target interactions in breast and liver cancer (Nerdy et al., 2021)	Vernodalin, Vernolide, Edotides, Luteolin, Quercetin	Apoptosis via mitochondrial pathway; immune activation via edotides; antioxidant NfκB modulation	Preclinical. Widely consumed as vegetable; generally recognized as safe (GRAS)
5.8	Aloe vera / Aloe africana Aloe Asphodelaceae	Aloe-emodin inhibits leukemia (HL-60) and HCC proliferation. Acemannan activates macrophage-mediated tumour killing. A. volkensii used in Kenyan multi-herb cancer preparations	Aloe-emodin, Barbaloin, Aloesin, Acemannan polysaccharide	Topoisomerase II inhibition; G2/M cell cycle arrest; immune-mediated tumour killing via macrophage activation	Aloe vera gel broadly sold. No cancer clinical trials. Used topically for radiation-induced dermatitis
5.9	Carica papaya Papaya Caricaceae	One of eight most-studied anticancer plants in Kenya. BITC from seeds suppresses STAT3 in breast cancer, induces apoptosis in HeLa and HCT116 cells; Lycopene associated with reduced prostate cancer risk in epidemiological studies	Benzyl isothiocyanate (BITC), Lycopene, Papain, Carpaine alkaloid, Quercetin	STAT3 signalling suppression; NfκB inhibition; caspase-3/9 activation; antioxidant carcinogen neutralisation	Preclinical evidence strong. Papain investigated as carrier for targeted drug delivery systems
5.10	Toddalia asiatica Orange Climber Rutaceae	Ranked among most-studied anticancer plants in Kenya. Root and leaf extracts cytotoxic against breast, cervical, colon cancer lines. Both alkaloid fractions and essential oil fractions active	Toddaline, Toddalinine (acridone alkaloids), Chelerythrine, Dictamnine, Isopimpinellin	Chelerythrine intercalates DNA; topoisomerase II inhibition; cell cycle arrest; intrinsic apoptosis	Preclinical. Widely used by East African traditional healers. Further toxicity profiling needed
5.11	Zanthoxylum chalybeum Knobwood Rutaceae	Alkaloid fractions active against leukemia and breast cancer cell lines in Ugandan research. Related species Z. paracanthum cited in Kenyan anticancer ethnobotany review (2022)	Benzophenanthridine alkaloids, Fagaridine, Chelerythrine, Nitidine	DNA intercalation; topoisomerase I/II inhibition; G1 and G2/M cell cycle arrest; apoptosis	Preclinical. Bark used by East African healers. HPLC profiling of alkaloids underway
5.12	Annona senegalensis Wild Soursop Annonaceae	N-hexane extract demonstrated antitumour activity in HCC model (N-diethylnitrosamine-induced). Augments liver architecture; upregulates p21 and p53 mRNA; downregulates AFP and farnesyl transferase	Alkaloids, Acetogenins, Flavonoids, Glycosides, Anthocyanins, Saponins	p53/p21 tumour suppressor upregulation; AFP downregulation; antioxidant enzyme enhancement; apoptosis	Preclinical in vivo (rat model) evidence. Important plant in Northern Nigeria. Further validation required
5.13	Cajanus cajan Pigeon Pea Fabaceae	Cajanol induces G2/M arrest in MCF-7 breast cancer cells via ROS-mediated mitochondrial pathway. IC50 in COR-L23 lung cancer cells: 5 to 10 ug/mL (stilbene fraction)	Cajanol (stilbene), Longstylin A and C (stilbenes), Cajanin, Genistein, Biochanin A	G2/M cell cycle arrest; ROS generation; cytochrome C release; caspase-3 and caspase-9 activation	Good preclinical data. Widely consumed as food crop globally (food-drug dual status noted)
5.14	Xylopia aethiopica Grains of Selim Annonaceae	Extracts induce MMP alterations (mitochondrial membrane potential disruption) in leukemia, colon, and breast cancer lines. Active in over 20 African countries. Included in Frontiers in Pharmacology CEWA plant review (Kuete et al., 2017)	Diterpenes, Ent-kaurane diterpenes, Xylopic acid, Essential oil (3-carene, terpinolene)	MMP disruption; intrinsic apoptosis; mitotic spindle interference; cytotoxic volatile compounds	Preclinical in vitro and in vivo evidence. Traditional use confirmed across 20+ African nations

No.	Species	Key Evidence	Primary Compounds	Mechanism	Status
5.15	<i>Morinda lucida</i> Brimstone Tree Rubiaceae	Anthraquinones including morindone, damnacanthal, and alizarin show antiproliferative activity against breast, colon, and HCC cell lines in Nigerian studies. Damnacanthal specifically inhibits tyrosine kinase (Lck/Src family)	Morindone, Damnacanthal, Alizarin, Rubiadin, Alkaloids	Topoisomerase II inhibition; tyrosine kinase inhibition (damnacanthal); apoptosis induction; oxidative stress	Preclinical. Well-established West African traditional use. Damnacanthal structure-activity studies ongoing
5.16	<i>Harungana madagascariensis</i> Dragon Blood Tree Hypericaceae	Used in Kenyan multi-herb preparations (with <i>Prunus africana</i>) against breast, skin, colorectal cancers. In vitro activity against breast and prostate cancer lines. Orange-red latex rich in bioactive phenolics	Harunganin, Madagascin, Anthraquinones (emodin-type), Xanthones	Cytotoxic anthraquinone and xanthone activity; apoptosis via mitochondrial pathway	Preclinical. Widely used by East African traditional healers. Combination preparations merit study
5.17	<i>Securidaca longipedunculata</i> Violet Tree Polygalaceae	Root extracts show cytotoxic activity against breast and cervical cancer lines. One of the most widely used medicinal plants in sub-Saharan Africa with over 40 documented uses	Methyl salicylate, Saponins (securidacoside series), Tannins, Cyanogenic glycosides	Cytotoxic saponin activity; DNA damage via saponin membrane disruption	Preclinical. Caution: roots are toxic at high doses. Therapeutic index research required
5.18	<i>Nymphaea lotus</i> Egyptian White Lotus Nymphaeaceae	Traditional use in Egypt, Uganda, Kenya for internal cancers. Preliminary in vitro evidence of antiproliferative activity in liver and breast cancer cell lines	Nuciferine, Nornuciferine, Quercetin, Kaempferol, Aporphine alkaloids	Antiproliferative alkaloid activity; aporphine alkaloid-mediated apoptosis; antioxidant flavonoid action	Early preclinical stage. Traditional use well-documented. Limited modern pharmacological data
5.19	<i>Alchornea cordifolia</i> Christmas Bush Euphorbiaceae	MMP alterations and apoptosis induction in leukemia and colon cancer lines confirmed in CEWA (Central, Eastern, Western Africa) plant review (Kuethe et al., 2017). Euphorbiaceae is the second most represented anticancer plant family in sub-Saharan Africa	Yohimbine-type alkaloids, Flavonoids, Tannins, Terpenoids, Alchorneinone	Mitochondrial apoptosis pathway; MMP disruption; cytotoxic alkaloid fraction activity	Preclinical in vitro. Traditional use documented across West and Central Africa
5.20	<i>Warburgia ugandensis</i> East African Greenheart Canellaceae	Warburganal and muzigadial are sesquiterpene dialdehydes showing cytotoxic activity against colon, breast, and leukemia cell lines. Related <i>W. salutaris</i> noted by Springer (2025) for antimicrobial and anticancer properties in Southern Africa	Warburganal (sesquiterpene dialdehyde), Muzigadial, Polygodial, Ugandensidial	Sesquiterpene-mediated cytotoxicity; protein synthesis inhibition; membrane disruption	Preclinical. Distinct phytochemical class not seen in most anticancer plants. Structure novel
5.21	<i>Ocimum gratissimum</i> African Basil Lamiaceae	Identified among most-studied anticancer plants in Kenya and Nigeria. Essential oil and leaf extracts show cytotoxic activity against MCF-7 breast, LNCaP prostate, and HeLa cervical cancer cell lines	Eugenol, Thymol, Rosmarinic acid, Ursolic acid, Beta-caryophyllene	NFκB pathway suppression; anti-inflammatory signalling reducing tumour promotion; apoptosis induction	Preclinical. Widely consumed as culinary herb. Excellent safety profile. Good GRAS status
5.22	<i>Peganum harmala</i> African Rue / Harmal Nitrariaceae	Harmine and harmaline are among the most chemically characterized anticancer alkaloids from North Africa. Active against lung (A549), colon (HCT116), breast (MCF-7), and leukemia cell lines. Harmine inhibits DYRK1A kinase, a target in glioblastoma and leukemia	Harmine (beta-carboline), Harmaline (dihydro-beta-carboline), Harman, Vasicine	DYRK1A kinase inhibition; topoisomerase I and II inhibition; CDK inhibition causing G1 arrest; MAO inhibition modifying tumour microenvironment	Well-characterized preclinical data. Harmine in early drug development pipeline for glioblastoma. North African traditional use well-documented
5.23	<i>Abelmoschus esculentus</i> Okra Malvaceae	Okra pectin fraction shown to inhibit colon carcinoma cell growth in murine models. Aqueous extracts demonstrate cytotoxic activity in breast and lung tumour lines. High dietary fibre content reduces colon carcinogen contact time (epidemiological evidence)	Quercetin-3-O-gentiobioside, Isoquercitrin, Hyperoside, Pectin, Polyphenols	Antiproliferative quercetin flavonoid action; pectin-mediated colon protection; antioxidant activity reducing DNA oxidative damage	Food-grade safety. Population-level protective evidence. Mechanistic studies ongoing. Least toxicologically problematic of all reviewed species

Table 3. Extended Species Profiles 5.7 through 5.23. MMP = Mitochondrial Membrane Potential; HCC = Hepatocellular Carcinoma; BITC = Benzyl Isothiocyanate; ROS = Reactive Oxygen Species; CDK = Cyclin-Dependent Kinase; NFκB = Nuclear Factor kappa B; GRAS = Generally Recognized as Safe.

6. MOLECULAR MECHANISMS OF ANTICANCER ACTION

The bioactive compounds identified from African medicinal plants operate through multiple distinct and often synergistic molecular mechanisms. Understanding these mechanisms is essential for rational drug development and for identifying which plants are most likely to yield successful clinical candidates.

Mechanism	Description	African Plant Examples	Compounds Involved
Tubulin Inhibition (Mitosis Arrest)	Binding to tubulin monomers prevents polymerization into microtubules, collapsing the mitotic spindle and arresting cell division at metaphase	Catharanthus roseus, Combretum cafrum, Xylopia aethiopica	Vinblastine, Vincristine, Combretastatin A-4, Diterpenes
Vascular Disruption (VDA Activity)	Selective collapse of immature tumour vasculature endothelial cells, cutting off blood supply and causing central tumour necrosis	Combretum cafrum	Combretastatin A-4 phosphate (Fosbretabulin)
Intrinsic Apoptosis (Mitochondrial Pathway)	Disruption of mitochondrial membrane potential (MMP), cytochrome C release, activation of caspase-9 and caspase-3, Bcl-2 downregulation	Annona muricata, Cajanus cajan, Sutherlandia frutescens, Vernonia amygdalina	Acetogenins, Cajanol, L-canavanine, Vernodalin
Topoisomerase I/II Inhibition	Prevents DNA unwinding required for replication; stabilizes DNA-enzyme cleavage complex causing permanent double-strand breaks, triggering apoptosis	Toddalia asiatica, Zanthoxylum spp., Aloe vera, Morinda lucida, Peganum harmala	Chelerythrine, Aloe-emodin, Damncanthal, Harmine, Nitidine
Cell Cycle Arrest	Blockade at G1, S, or G2/M checkpoints; prevents progression through cell cycle; forces senescence or apoptosis in tumour cells	Cajanus cajan (G2/M), Peganum harmala (G1), Aloe vera (G2/M)	Cajanol (ROS-mediated), Harmine (CDK inhibition), Aloe-emodin
NFκB Pathway Suppression	Inhibits nuclear factor kappa B transcription factor, reducing expression of anti-apoptotic, pro-inflammatory, and pro-angiogenic genes in tumour cells	Moringa oleifera, Ocimum gratissimum, Annona muricata, Carica papaya	Isothiocyanates, Eugenol, Acetogenins, BITC
Tumour Suppressor Gene Upregulation	Upregulation of p53, p21, and Bax while downregulating anti-apoptotic Bcl-2/Bcl-xL restores apoptotic competence to cancer cells	Annona senegalensis, Moringa oleifera, Annona muricata	n-hexane acetogenin extract, Isothiocyanates, Flavonoids
Androgen Receptor Antagonism	Blocking AR nuclear translocation suppresses androgen-responsive gene expression; mimics pharmaceutical anti-androgen therapy in prostate cancer	Prunus africana	N-butylbenzene sulfonamide (NBBS), Atraric acid
Immunostimulation (Indirect)	Activation of macrophages, natural killer cells, and T-lymphocytes to enhance natural tumour immunosurveillance and killing	Sutherlandia frutescens, Aloe vera, Vernonia amygdalina	D-pinitol, Acemannan polysaccharide, Edotides
Anti-Angiogenesis (VEGF Inhibition)	Suppression of vascular endothelial growth factor (VEGF) reduces tumour-induced neovascularization, starving tumours of nutrient and oxygen supply	Moringa oleifera, Carica papaya	Quercetin glycosides, Benzyl isothiocyanate (BITC)

Table 4. Molecular Mechanisms of Anticancer Action Documented in African Plant-Derived Compounds. VDA = Vascular Disrupting Agent; MMP = Mitochondrial Membrane Potential; CDK = Cyclin-Dependent Kinase; VEGF = Vascular Endothelial Growth Factor; NFκB = Nuclear Factor Kappa B; AR = Androgen Receptor.

7. GEOGRAPHIC AND TAXONOMIC DISTRIBUTION

Anticancer medicinal plant research in Africa spans all major ecological zones. The distribution of both plant diversity and research activity is uneven, reflecting funding concentrations and academic infrastructure. Sub-Saharan Africa, particularly East and West Africa, accounts for the majority of documented species.

Region	Countries with Most Documentation	Dominant Plant Families	Representative Species	No. of Reviewed Species
East Africa	Kenya, Uganda, Tanzania, Ethiopia	Asteraceae, Euphorbiaceae, Rosaceae, Rutaceae, Annonaceae	P. africana, T. asiatica, Z. chalybeum, W. ugandensis	8
West Africa	Nigeria, Ghana, Cameroon, Senegal, Ivory Coast	Euphorbiaceae, Rubiaceae, Annonaceae, Fabaceae, Asteraceae	A. muricata, V. amygdalina, M. lucida, A. senegalensis	7
Southern Africa	South Africa, Zimbabwe, Botswana, Zambia, Namibia	Fabaceae, Combretaceae, Asphodelaceae, Hypericaceae	S. frutescens, C. caffrum, Aloe spp.	4
Pan-Africa (Multiple Regions)	Widespread across Sub-Saharan Africa	Moringaceae, Caricaceae, Lamiaceae, Malvaceae	M. oleifera, C. papaya, O. gratissimum, A. esculentus	4
North Africa	Morocco, Algeria, Tunisia, Libya, Egypt	Nitrariaceae, Nymphaeaceae, Apiaceae, Ranunculaceae	P. harmala, N. lotus	2
Madagascar (Island)	Madagascar	Apocynaceae, Hypericaceae	C. roseus (origin)	1 (origin)

Table 5. Geographic Distribution and Taxonomic Composition of Reviewed African Anticancer Plant Species.

The Asteraceae and Euphorbiaceae families are consistently the most represented in East African anticancer ethnobotany. The Annonaceae family (genus Annona and related) is notably productive in West and Central Africa, contributing acetogenins as a structurally unique class of anticancer agents. The Fabaceae family is the most geographically distributed, present across all African regions with documented anticancer species (Sutherlandia, Cajanus, Albizia).

8. CHALLENGES, LIMITATIONS, AND RESEARCH GAPS

Absence of Clinical Trials

With the exception of vinca alkaloids (clinically approved, 1960s) and fosbretabulin (Phase II/III), virtually no African plant-derived anticancer compound has completed rigorous Phase I through Phase III human clinical trials. The overwhelming majority of evidence remains at in vitro cell line or small animal model stage. Robust efficacy and safety data in human patients is lacking for all but the most advanced compounds.

Standardization and Batch Variability

Plant-derived medicines are chemically heterogeneous. Identical species collected from different geographic locations, soil types, seasons, altitude, or developmental stages may contain substantially different concentrations of active compounds. A study on *Sutherlandia frutescens* by Zonyane et al. (2020) confirmed considerable qualitative and quantitative chemical variability across South African populations, directly affecting bioactivity outcomes.

Toxicity and Adverse Effect Profiles

Several promising plants carry documented safety concerns. *Annona muricata* acetogenins have been epidemiologically linked to atypical Parkinsonism in long-term consumers in French Caribbean populations. L-canavanine in *Sutherlandia frutescens* may aggravate systemic lupus erythematosus and autoimmune disorders. *Securidaca longipedunculata* roots are toxic at near-therapeutic doses. Thorough toxicological profiling is absent for most reviewed species.

Ethnobotanical Misidentification

Traditional healers may use similar vernacular names for botanically distinct species across regions. Specimen voucher deposition, DNA barcoding, and formal herbarium verification are inconsistently applied in African ethnobotanical studies, creating reproducibility challenges.

Funding and Infrastructure Deficits

Less than 1% of global cancer research funding reaches sub-Saharan African institutions. Most rigorous anticancer plant pharmacology studies are conducted by research groups in Europe, North America, or Asia using African plant extracts, often without adequate co-authorship of African scientists or benefit-sharing with source communities. Local capacity for clinical-grade pharmacological research remains severely underdeveloped.

Conservation and Sustainability

Commercial demand for bark of *Prunus africana* has led to overexploitation and CITES Appendix II listing. Multiple other high-demand medicinal trees face localized extinction pressure. As deforestation continues across equatorial Africa, the race to document and conserve medicinal plant diversity before it is lost is urgent and inadequately resourced.

Intellectual Property and Biopiracy Risk

Traditional knowledge of plant medicines belongs to African communities under CBD/Nagoya Protocol provisions. Historical biopiracy — the commercial exploitation of African plant knowledge without benefit-sharing — remains an ethical and legal issue. The vinblastine/vincristine case from Madagascar generated billions in pharmaceutical revenues with negligible benefit return to the source country.

9. CONCLUSION AND RECOMMENDATIONS

Africa is home to an extraordinary and largely undercharacterized wealth of medicinal plants. The continent has already made irreplaceable contributions to global oncology pharmacology through *Catharanthus roseus* (vinca alkaloids, FDA-approved, WHO Essential Medicines) and *Combretum caffrum* (combretastatins, Phase III trials). The 23 additional species reviewed here collectively demonstrate a substantial body of preclinical anticancer evidence spanning apoptosis induction, cell cycle arrest, topoisomerase inhibition, NFκB suppression, androgen receptor antagonism, vascular disruption, and immunostimulation.

Research has been initiated by a multi-team collaborative at Zetech University, Kenya, recognizing that the translation of this preclinical evidence into clinical benefit for African cancer patients requires a structured, systematic, and ethically grounded research programme. The ZTUMTRI-2025 initiative represents a commitment by Zetech University to contribute to this national and continental research priority.

9.1 Priority Species for Advanced Investigation

Based on the combined weight of preclinical evidence, degree of molecular characterization, and safety profiles, the following species are recommended as highest priority for advancement to clinical-stage investigation:

- **Combretum caffrum:** Fosbretabulin (CA4P) already in Phase III — continued trial support and expansion to additional solid tumour indications
- **Sutherlandia frutescens:** Phase I safety/PK study with standardized SU1 extract in Southern African patients; autoimmune exclusion criteria essential
- **Annona muricata:** Purified acetogenin fraction with safety monitoring for neurological adverse events; nanoparticle delivery platforms to reduce toxicity
- **Moringa oleifera:** Phase I ITC-enriched extract in breast and colon cancer; excellent safety profile and food-grade precedent support rapid clinical entry
- **Peganum harmala (Harmine):** Harmine in early pipeline for glioblastoma; North African clinical site development recommended
- **Prunus africana:** Standardized bark extract in prostate cancer prevention trial; leverage existing European BPH phytomedicine regulatory pathway

9.2 Structural Recommendations

- Establish a Pan-African Medicinal Plant Anticancer Research Consortium coordinated through the African Union
- Mandate voucher specimen deposition and DNA barcoding for all future African ethnopharmacology publications
- Develop Good Agricultural and Collection Practices (GACP) guidelines for priority species to ensure extract consistency
- Create equitable benefit-sharing mechanisms linking pharmaceutical development revenues to source communities and national biodiversity funds
- Integrate traditional healer knowledge formally into academic research teams with credited authorship and fair compensation
- Establish cultivation programmes for threatened species including *Prunus africana* and *Warburgia ugandensis*

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The following peer-reviewed sources were reviewed, cited, and used as evidence base for this report. Full bibliographic details are provided for academic citation verification.

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END OF REPORT — ZETECH UNIVERSITY MULTI-TEAM RESEARCH INITIATIVE (ZTUMTRI-2025)

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This report does not constitute medical advice. No herbal preparation reviewed herein has been approved as a standalone cancer cure. Patients should consult qualified oncologists.